

Milestone #2
Semipalmated Sandpiper (SESA) Migration Stopover Site Analysis
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Handle Missing Values

Only one of my datasets had missing values, and that was my ebird dataset. These missing values occurred in the columns duration_minutes, effort_distance_km, number_observers, and observation_count_clean. I imputed the median for three of the columns (columns duration_minutes, effort_distance_km, number_observers) because they were right skewed with outliers so the median was less biased than the mean to use. I filled observation_count_clean with 0 because SESA was recorded as present at that site, but the count was not recorded, missing count does not translate to no SESA in that location. My other datasets did not have missing values, and no missing values were in columns that will be used directly for model prediction.

Address Data Imbalance

There is a slight data imbalance in my data, there are 200 presences that will be considered as hotspots for my analysis, and there are 500 pseudo-absences. The pseudo-absence approach was decided on because it is a standard practice in species distribution modeling and it is interpretable. Random points represent these absence sites, they are not confirmed absences. Regarding metrics and evaluation - using accuracy alone would be misleading due to the class imbalance. Instead a combination of AUC-ROC, precision, recall, and f1-score will be used to evaluate.

Feature Engineering

One hot encoding was applied to land cover data, to assign categorical features such as land cover classes to binary 0 and 1 labels. Habitat data was scaled in order to be used in linear models, specifically n_detections and n_years. Three binary columns were created: land_cover_changed, habitat_degraded, habitat_improved.. These noted habitat change, and whether or not the habitat was suitable and became not suitable, or vice versa. Good habitat classes included the following: Emergent Herbaceous Wetlands, Open Water, Woody Wetlands, Grassland/Herbaceous, Pasture/Hay and these were determined based on prior knowledge of SESA habitat characteristics. Both slr_05m and slr_10m were dropped, this is because all sites are inundated in both scenarios so they provide no predictive power for the model, however that finding is still interesting. (I may in the future alter the sea level rise scenarios to see if there is a scenario where only some sites are inundated.) Both nlcd_2001 and seasons were dropped as well, nlcd_2001 was no longer needed after the change columns with binary labels were created, and temporal information can be obtained from the ebird dataset and is therefore not needed in the habitat dataset. The final dataset that will be used for modeling is 700 rows - comprised of 200 presences and 500 pseudo absences.

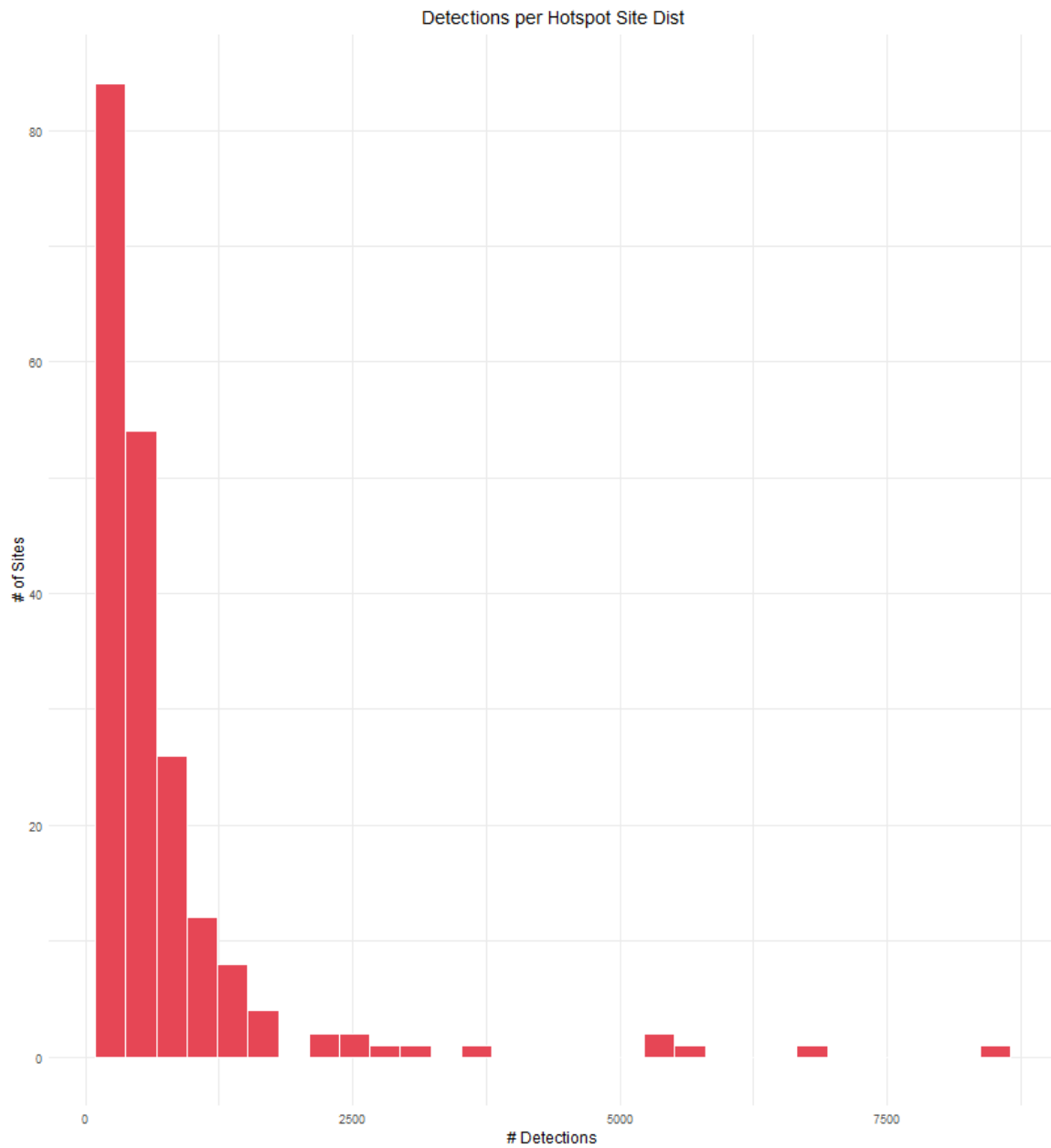
Outlier Detection

Outliers were detected in n_detections as well as n_years. They were not dropped as they do represent stopover sites that are extremely heavily used by SESA and are not data errors. Boxplots showed visual confirmation of the right skew that was present in n_detections.

Summary Statistics

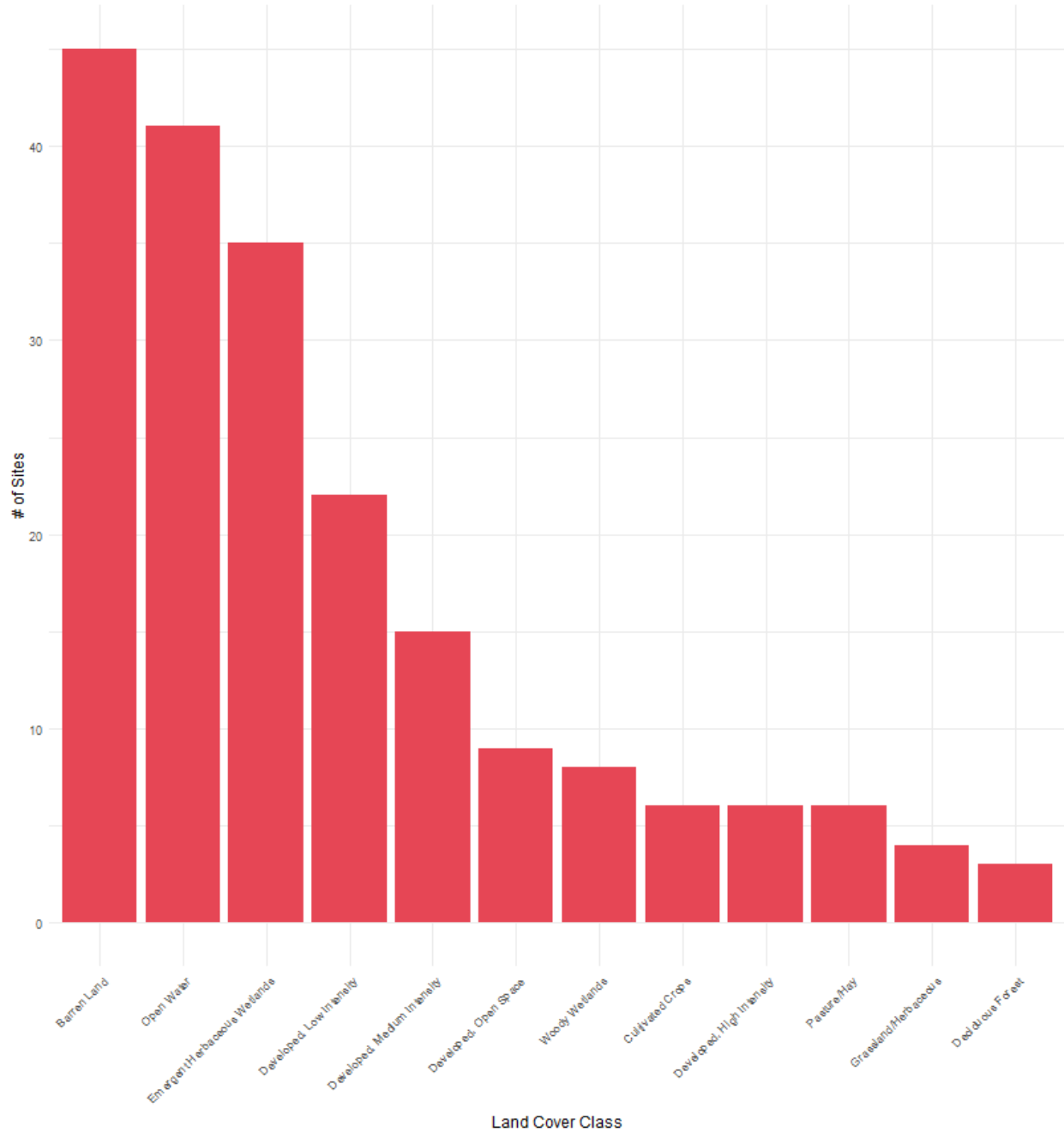
n_detections had a wide range as discussed in outlier section above, majority of hotspots had typical amounts of detections but there was a small number of hotspots that were those high used sites that had a large increase in detections. N_years had a range from 3-25, majority of hotspots had lengthy detection histories going back many years which supports the idea of hotspots that have been consistently used by SESA over many years.

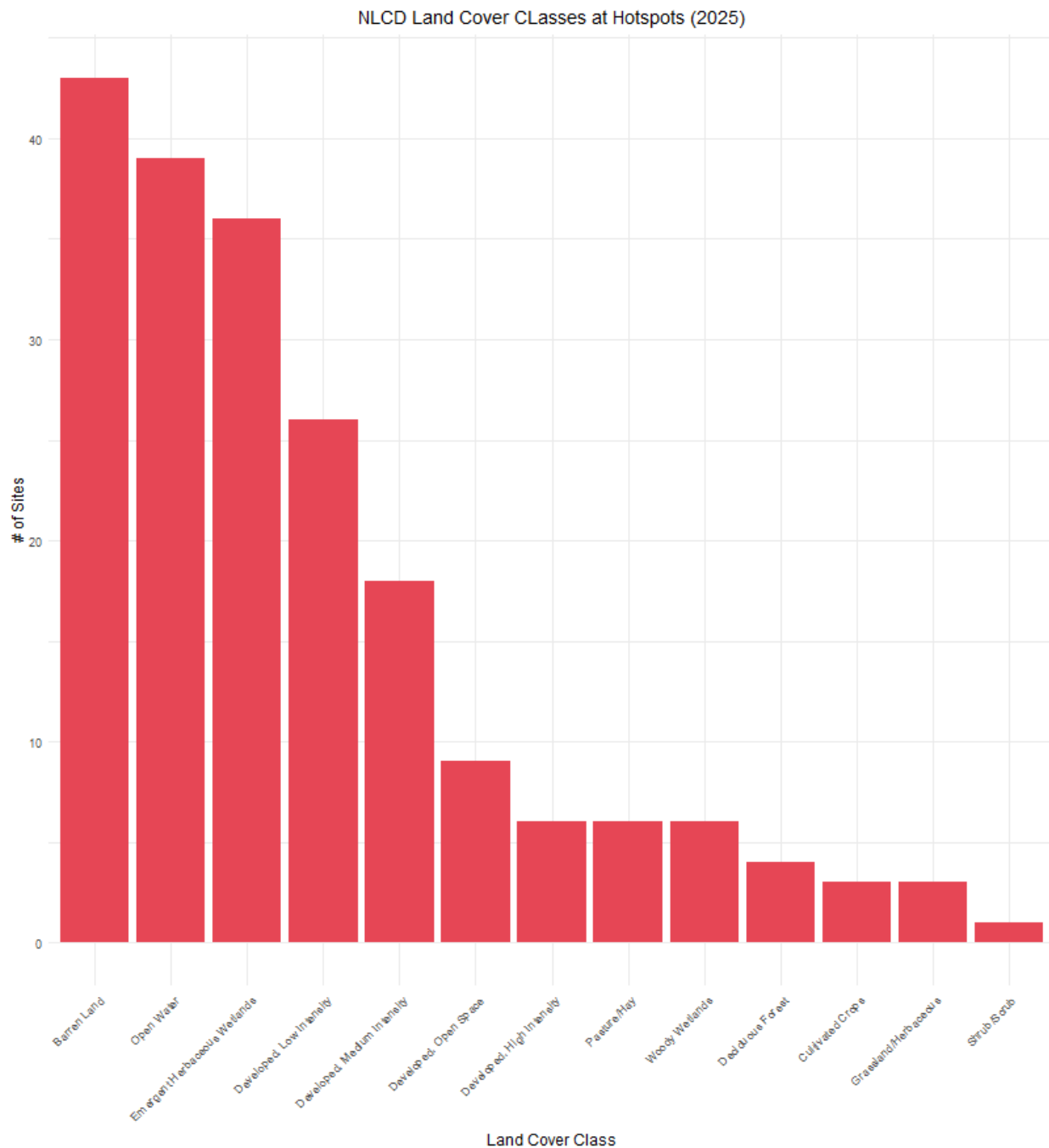
Visualizations



Detections per Hotspot Site Dist: right skewed, shows small number of highly utilized hotspots and large number of moderately used hotspots

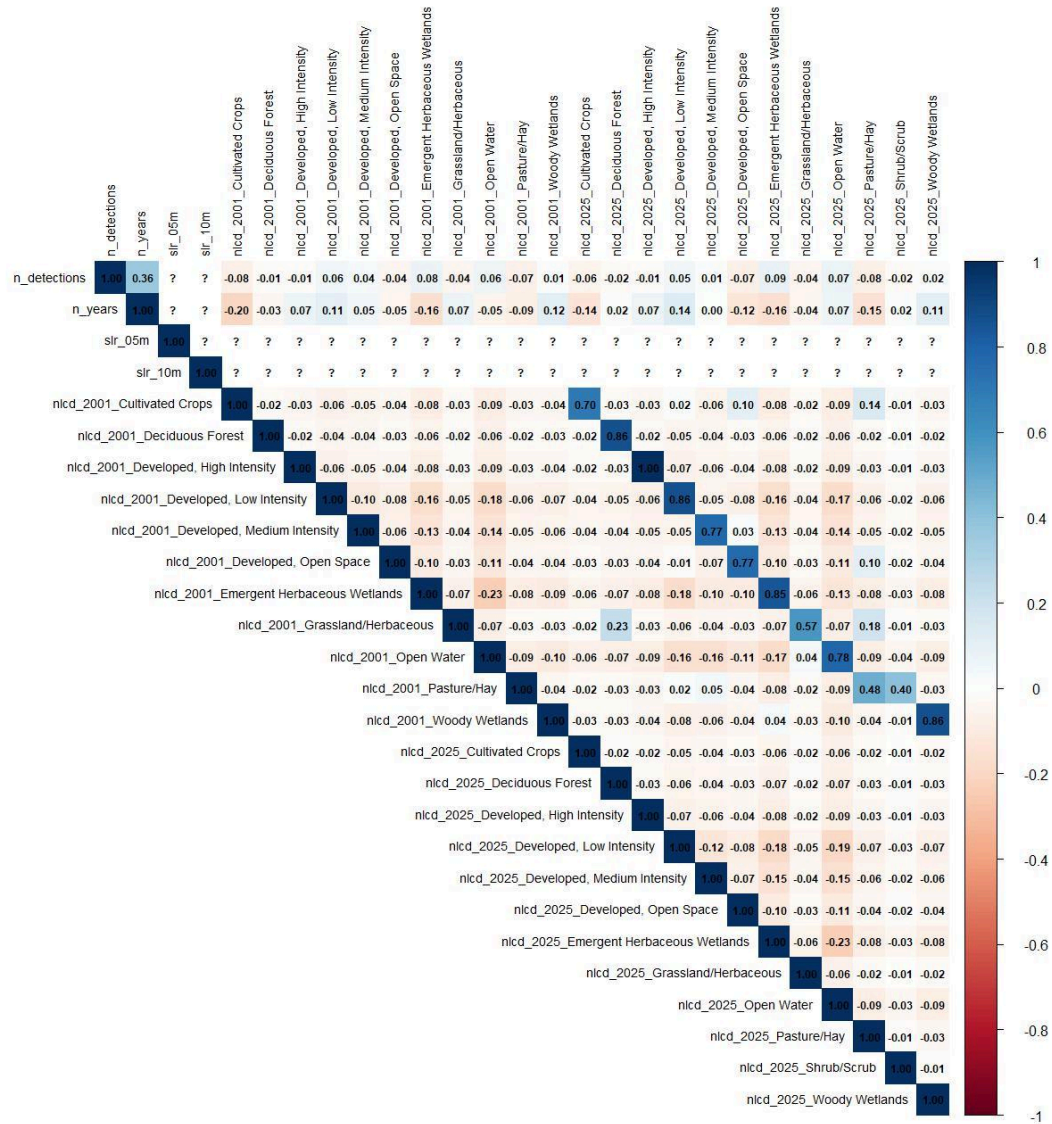
NLCD Land Cover CLASSES at Hotspots (2001)



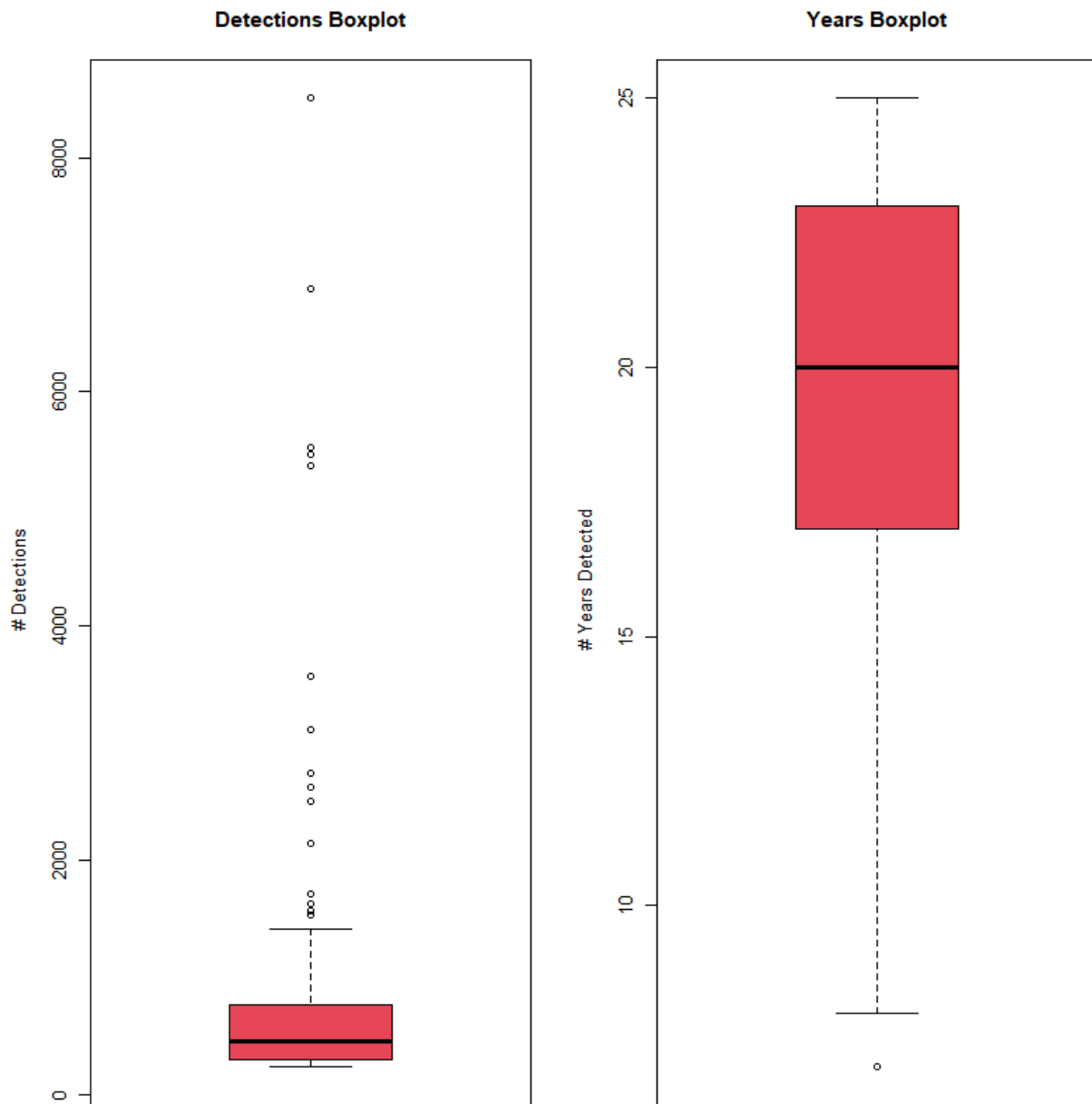


NLCD Land Cover Classes at Hotspots: barren land, open water, and emergent herbaceous wetlands dominate the land cover classes as stopover sites - this is consistent from 2001 to 2025.

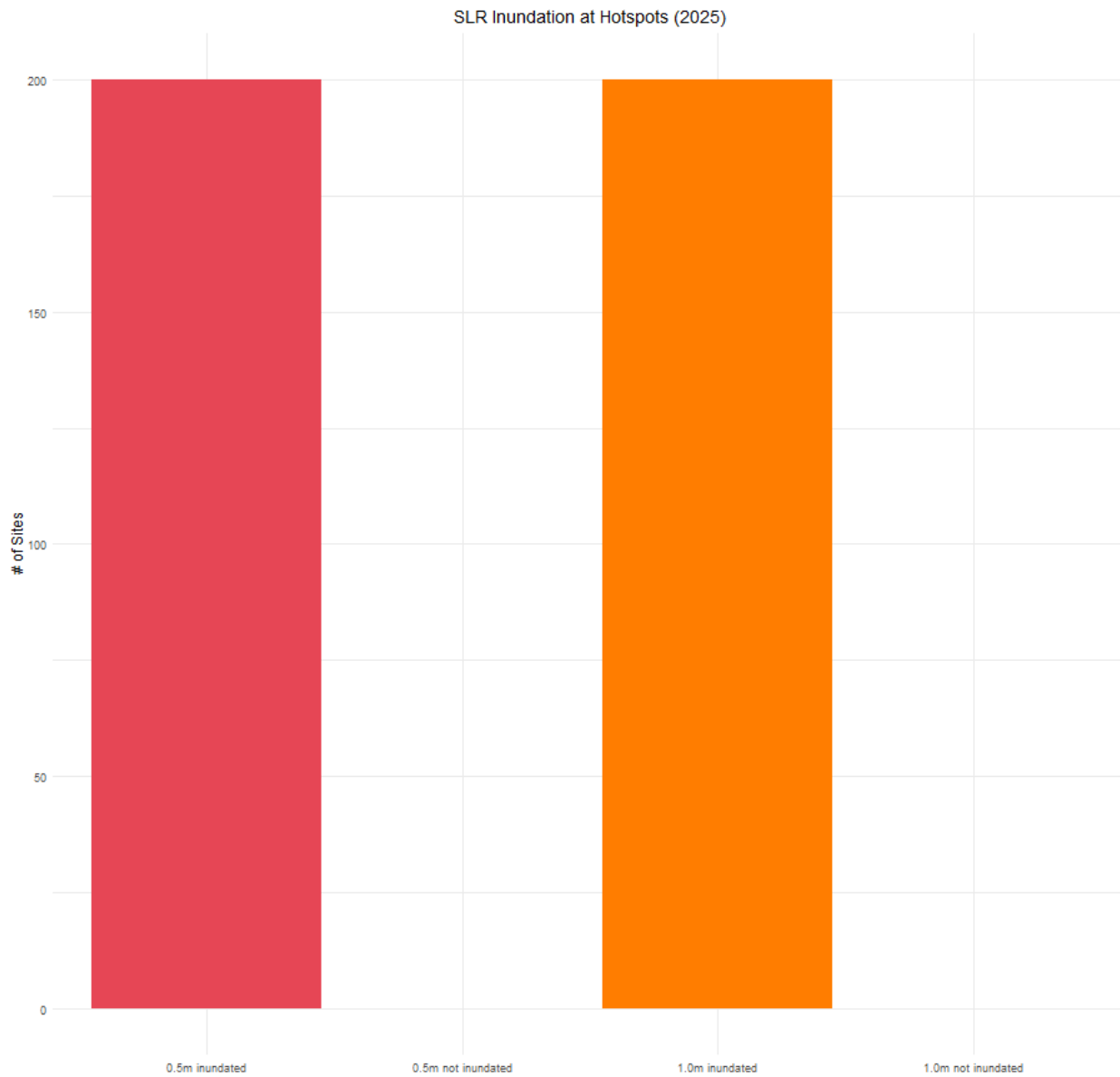
Feature Correlation Heatmap



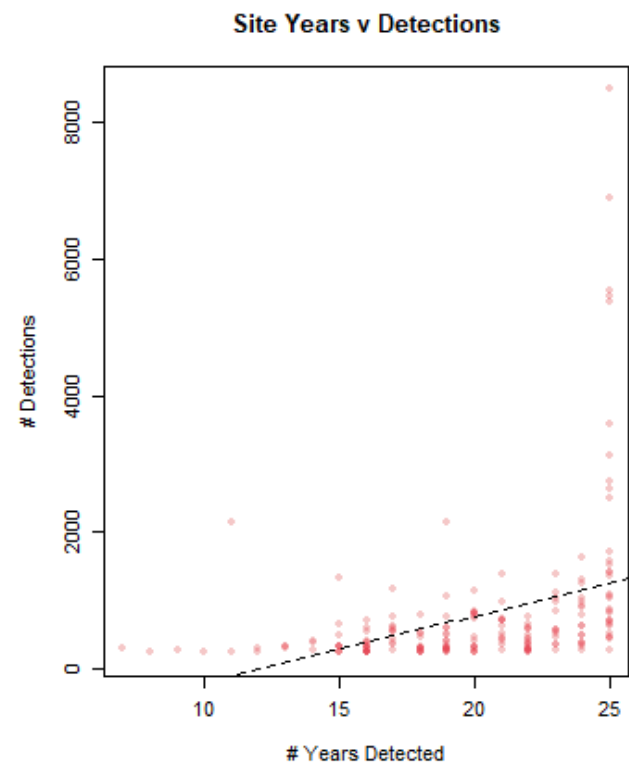
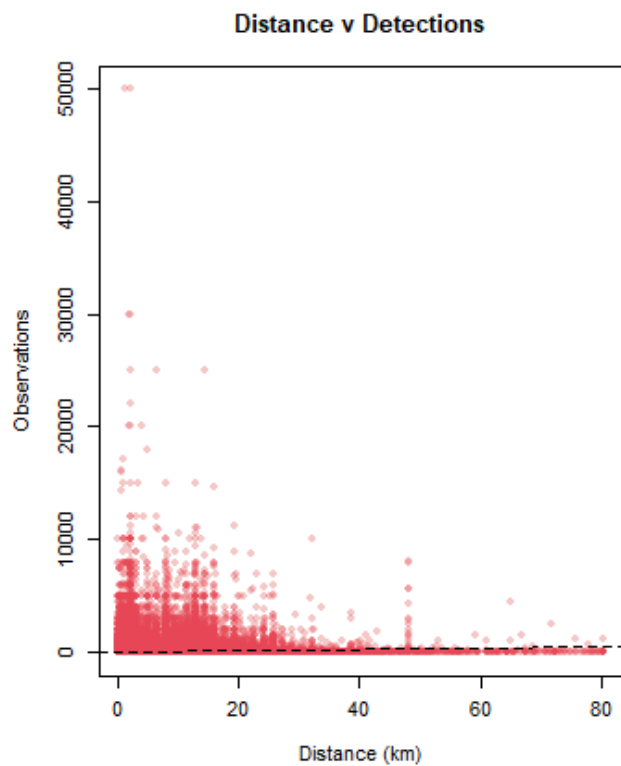
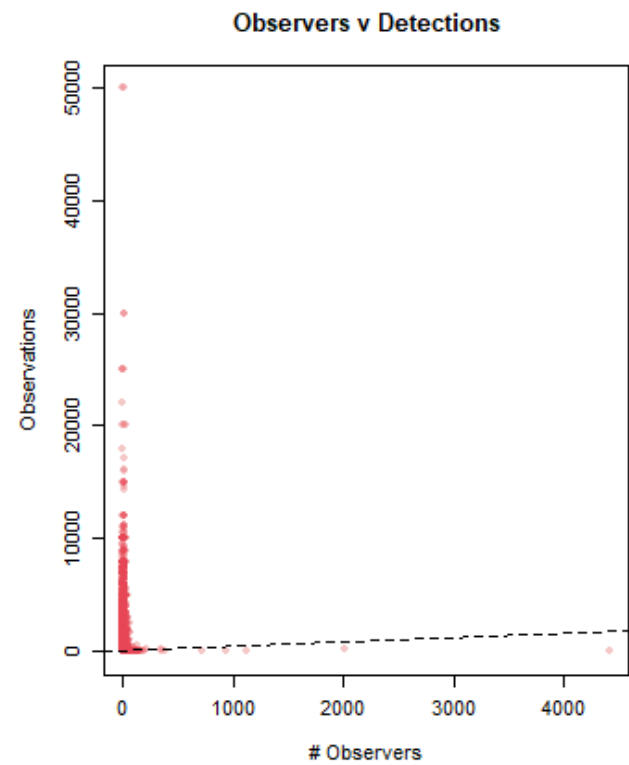
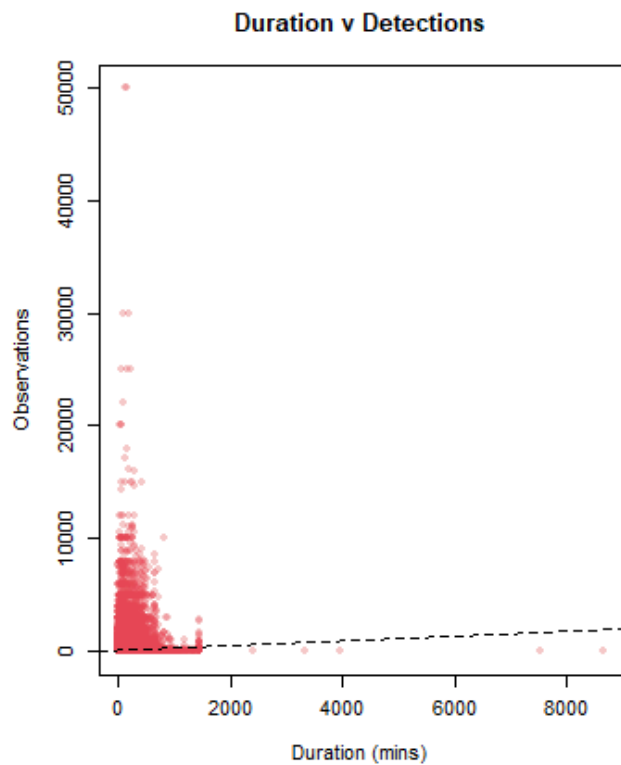
Feature correlation heatmap: shows some high correlation between features, especially between OHE land cover columns.



Detection and Year outliers: Some extreme outliers seen in detections, due to previously discussed concepts, not many outliers in years.



Sea level rise inundation at hotspots. This shows the two sea level rise scenarios I wanted to evaluate, both scenarios completely inundate all hotspots in my analysis.



Scatter plots show weak linear relationships between distance/duration/observers and detections. Scatter plot between site years and detections shows a strong linear relationship between the two variables.

Patterns Tied to Hypothesis

Through EDA a few patterns have been uncovered so far. NLCD classes as hotspots are in line with the habitat preferences of SESA which supports my hypothesis that land cover is a meaningful predictor of hotspots. SESA is sensitive to / seen often in vulnerable coastal environments and those environments have more drastic habitat changes. There are many years worth of detections at most of the hotspots which supports n_years being used as an indicator for the quality of the site, sites used year after year must have good resources and shelter, etc that is beneficial for migration. Sea level rise (0.5 and 1.0m) inundation at all sites serves no beneficial modeling purpose but is in itself an interesting finding. Long term stopover sites are at severe risk.